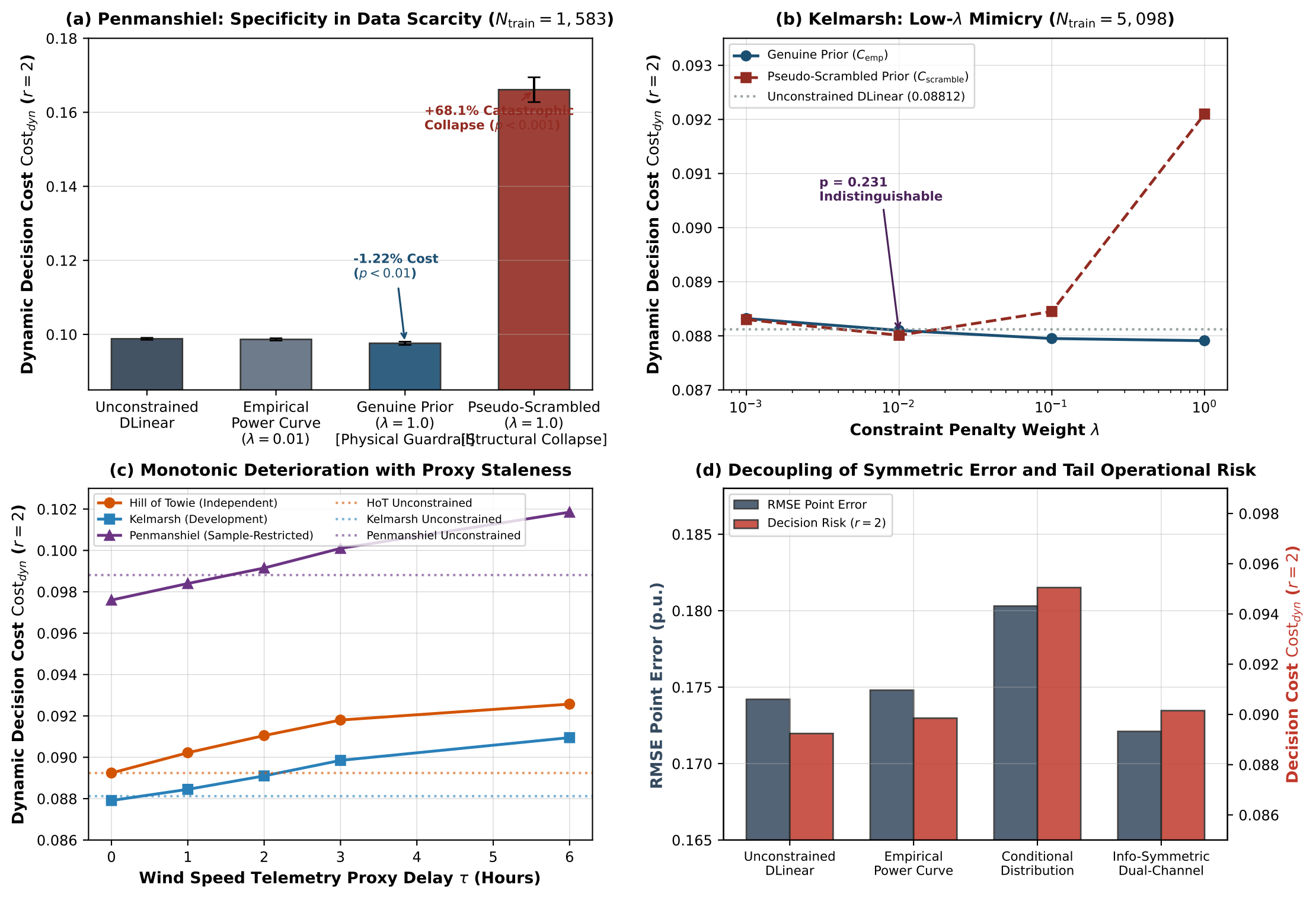

(a) Penmanshiel: Specificity in Data Scarcity ($N_{\text{train}} = 1,583$)
Dynamic Decision Cost Cost_{dyn} ($r = 2$)
0.18
0.16
0.14
0.12
0.10
+68.1% Catastrophic Collapse ($p < 0.001$)
-1.22% Cost ($p < 0.01$)
Unconstrained DLinear
Empirical Power Curve ($\lambda = 0.01$)
Genuine Prior ($\lambda = 1.0$) [Physical Guardrails]
Pseudo-Scrambled ($\lambda = 1.0$) [Structural Collapse]
(b) Kelmarsh: Low-λ Mimicry ($N_{\text{train}} = 5,098$)
Genuine Prior (C_{emp})
Pseudo-Scrambled Prior (C_{scramble})
Unconstrained DLinear (0.08812)
p = 0.231 Indistinguishable
Dynamic Decision Cost Cost_{dyn} ($r = 2$)
0.093
0.092
0.091
0.090
0.089
0.088
0.087
10^{-3}
10^{-2}
10^{-1}
10^{0}
Constraint Penalty Weight λ
(c) Monotonic Deterioration with Proxy Staleness
Hill of Towie (Independent)
Kelmarsh (Development)
Penmanshiel (Sample-Restricted)
HoT Unconstrained
Kelmarsh Unconstrained
Penmanshiel Unconstrained
Dynamic Decision Cost Cost_{dyn} ($r = 2$)
0.102
0.100
0.098
0.096
0.094
0.092
0.090
0.088
0.086
0
1
2
3
4
5
6
Wind Speed Telemetry Proxy Delay τ (Hours)
(d) Decoupling of Symmetric Error and Tail Operational Risk
RMSE Point Error
Decision Risk ($r = 2$)
RMSE Point Error (p.u.)
0.185
0.180
0.175
0.170
0.165
Decision Cost Cost_{dyn} ($r = 2$)
0.098
0.096
0.094
0.092
0.090
0.088
0.086
Unconstrained DLinear
Empirical Power Curve
Conditional Distribution
Info-Symmetric Dual-Channel